

ABHA, Saudi Arabia

WMO: 411120

Lat: 18.240N

Lon: 42.657E

Elev: 6858

StdP: 11.40

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 44.3	(c) 46.3	(d) 8.8	(e) 11.2	(f) 60.2	(g) 14.2	(h) 14.6	(i) 64.8	(j) 23.3	(k) 62.0	(l) 21.4	(m) 62.3	(n) 1.7	(o) 330	(p) 0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 22.3	(c) 89.6	(d) 56.6	(e) 88.0	(f) 56.2	(g) 86.4	(h) 56.3	(i) 68.2	(j) 76.1	(k) 67.1	(l) 75.1	(m) 66.0	(n) 74.1	(o) 12.0	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	124.7	72.1	64.6	118.2	71.7	63.0	111.6	70.9	37.2	76.0	36.2	75.0	35.1	74.1	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.5	16.7	38.5	92.2	2.6	2.0	36.7	93.6	35.2	94.8	33.7	95.9	31.8	97.3
			29.8	70.5	2.7	2.0	27.9	71.9	26.3	73.1	24.8	74.2	22.9	75.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	57.5	60.6	63.7	67.3	72.0	75.9	75.9	75.1	73.0	66.9	61.6	58.2		
	DBStd	7.24	2.91	2.99	3.30	3.14	3.18	2.40	2.84	2.52	2.53	2.62	2.63	2.42		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	765	234	126	63	14	1	0	0	0	0	11	105	211		
	CDD50	6328	232	297	424	518	682	776	803	779	691	524	348	254		
	CDD65	1617	0	2	22	82	219	326	338	314	241	70	3	0		
	CDH74	12663	9	33	152	525	1671	2834	2598	2254	1985	548	48	6		
	CDH80	3754	0	0	5	43	430	1037	890	765	550	34	0	0		
(14) Wind	WSAvg	6.9	7.9	8.4	9.1	7.5	6.1	6.5	6.8	6.6	6.7	5.8	5.6	6.2		
(15) Precipitation	PrecAvg	7.5	0.4	0.4	1.1	1.7	1.1	0.3	0.7	1.0	0.3	0.1	0.2	0.1		
	PrecMax	19.7	2.0	3.9	13.5	8.8	3.9	2.6	3.0	4.8	1.5	0.9	1.5	0.7		
	PrecMin	1.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0		
	PrecStd	4.2	0.6	0.8	2.6	2.2	1.0	0.5	0.6	0.9	0.4	0.3	0.4	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	77.2	80.5	83.7	89.2	91.5	91.7	90.0	89.5	82.7	77.4	74.8		
		MCWB	51.2	53.6	55.6	56.6	56.6	57.0	57.6	57.9	56.3	53.5	53.4	49.5		
	2%	DB	71.9	75.0	77.4	80.9	86.4	89.4	89.4	88.1	87.1	80.7	75.3	71.8		
		MCWB	52.5	54.3	55.2	56.4	56.0	55.8	56.8	57.1	55.7	52.5	53.8	52.2		
	5%	DB	69.6	73.0	75.4	79.1	84.4	87.7	87.5	86.4	85.3	79.0	73.5	69.9		
		MCWB	52.7	54.5	55.6	57.0	56.7	55.3	56.8	57.3	55.2	52.2	53.7	52.7		
	10%	DB	66.7	70.2	73.4	77.1	82.4	86.0	85.1	84.5	83.3	77.2	71.6	68.0		
		MCWB	53.1	54.3	56.1	57.5	57.3	55.5	57.5	58.1	54.7	52.2	53.9	53.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.6	61.2	63.6	64.7	67.9	68.5	69.3	69.9	67.9	64.2	62.8	60.2		
		MCDB	67.1	68.9	71.0	72.4	75.7	77.2	77.5	77.3	75.1	69.3	68.7	66.7		
	2%	WB	58.1	59.9	61.6	63.2	66.0	67.0	67.5	68.3	66.0	62.6	61.1	58.8		
		MCDB	65.3	67.7	69.6	71.9	74.3	76.0	76.1	75.9	73.1	68.6	67.1	65.5		
	5%	WB	56.9	58.5	60.4	62.1	64.6	65.4	66.2	67.2	64.5	61.1	60.0	57.7		
		MCDB	63.5	66.4	68.6	71.4	73.2	75.2	74.6	74.8	71.6	67.3	66.0	64.1		
	10%	WB	56.0	57.5	59.4	61.0	63.0	63.5	64.8	66.0	62.6	59.7	58.8	56.6		
		MCDB	62.2	65.1	67.7	70.2	73.1	74.5	73.5	73.6	71.5	66.6	65.0	62.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	19.6	18.7	19.9	21.5	22.3	20.7	21.5	22.6	22.5	21.2	21.0		
		MCDBR	24.8	23.7	22.3	22.2	23.3	23.7	22.4	23.0	23.5	23.5	23.6	24.4		
		MCWBR	15.4	14.5	12.6	10.7	12.0	13.4	11.8	12.4	13.1	13.5	13.1	14.4		
	5% WB	MCDBR	19.3	20.3	18.8	19.7	21.7	22.7	20.8	21.6	22.6	22.5	20.9	20.2		
		MCWBR	12.2	13.0	11.0	10.5	12.3	14.8	12.8	12.7	15.0	15.0	11.9	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.330	0.367	0.400	0.441	0.500	0.556	0.615	0.602	0.499	0.392	0.370	0.335		
	taud		2.418	2.276	2.168	2.057	1.876	1.718	1.637	1.662	1.860	2.168	2.261	2.386		
	Ebn at Noon		303	296	288	276	257	240	227	232	256	283	286	297		
	Edh at Noon		35	42	49	55	65	75	82	80	65	46	40	35		
(44) All-Sky Solar Radiation	RadAvg		1606	1844	2085	2186	2281	2294	2086	2010	2081	2012	1701	1572		
	RadStd		86	94	97	119	112	77	88	65	61	69	59	88		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.13	+0.88	N/A	+1.64	N/A	N/A	N/A	-189	+499	+311
Regional (4 neighbors)	+0.85	N/A	N/A	+0.88	N/A	N/A	N/A	-20	+346	+277

Nomenclature: See separate page